Keldysh–Floquet Formulation of Voltage- and Current-Biased Josephson Junctions

Technical notes for the `Keldyshsetup_Floquetn` codebase

Abstract

These notes give a self-contained derivation of the equations solved by the Julia codebase that accompanies the paper *"Origin of Subharmonic Gap Structure of DC Current-Biased Josephson Junctions"* (arXiv:2412.09862). We set up the nonequilibrium (Keldysh) Green's-function description of a single-channel Josephson junction of arbitrary transparency in Nambu space, introduce the Floquet (harmonic) representation generated by the time-dependent superconducting phase, and write the Dyson, Keldysh and current equations. We then specialize to the two driving protocols: *voltage bias* (constant V, a single-shot evaluation) and *current bias* (constant time-averaged voltage $\langle V \rangle$, requiring a self-consistent solution for the phase dynamics). Throughout we use units $\hbar = e = 1$ and indicate the correspondence to the routines in `Keldyshsetup_Floquetn.jl`.

Contents

1 Physical setup

We model a Josephson junction as two BCS superconducting leads (left L and right R) coupled through a single conduction channel with hopping (tunnelling) amplitude $\mathcal{T}$. Each lead is a semi-infinite system characterized by:

- the superconducting gap Δ (all energies are measured in units of Δ; the code sets `delta` $= 1$),
- an internal hopping / bandwidth scale ζ (`zeta`), which controls the normal-state lead coupling and hence the normalization of the surface Green's function,
- a phenomenological Dynes-type broadening Γ (`Gamma`) that regularizes the BCS singularities and supplies the lesser (occupation) self-energy through a coupling to equilibrium reservoirs.

The single dimensionless parameter that controls the junction transparency is the ratio $\mathcal{T}/\zeta$: $\mathcal{T}/\zeta \ll 1$ is the tunnel limit, while $\mathcal{T}/\zeta \to 1$ approaches the high-transparency (ballistic) regime. The chemical potential is $\mu = 0$ (particle–hole symmetric point).

1.1 Hamiltonian and the superconducting phase

In Nambu (particle–hole) space we use the spinor $\Psi_\alpha = (c_{\alpha\uparrow}, c^\dagger_{\alpha\downarrow})^\mathsf{T}$, $\alpha \in \{L, R\}$, and the Pauli matrices $\tau_0, \tau_1, \tau_2, \tau_3$ acting on this space, with $\tau_3 = \mathrm{diag}(1, -1)$. Each lead is described by a mean-field BCS Hamiltonian with order parameter $\Delta\, \mathrm{e}^{\mathrm{i}\varphi_\alpha}$. The junction Hamiltonian is

$$H(t) = H_L + H_R + H_T(t), \tag{1}$$

where the coupling term, after a gauge transformation that moves the entire phase difference $\phi(t) = \varphi_L - \varphi_R$ onto the hopping, reads

$$H_T(t) = \mathcal{T}\, \Psi_L^\dagger \begin{pmatrix} \mathrm{e}^{-\mathrm{i}\phi(t)/2} & 0 \\ 0 & -\,\mathrm{e}^{+\mathrm{i}\phi(t)/2} \end{pmatrix} \Psi_R + \mathrm{h.c.} \tag{2}$$

The two Nambu entries carry opposite phase factors $\mathrm{e}^{\mp\mathrm{i}\phi/2}$ because the electron and hole components couple to the pairing field with opposite charge.

1.2 Josephson relation and the two bias protocols

The phase difference is tied to the voltage drop across the junction by the Josephson (gauge) relation

$$\dot{\phi}(t) = 2\,V(t). \tag{3}$$

Two distinct steady-state driving protocols follow:

Voltage bias. $V(t) = V_{\mathrm{dc}}$ is held constant. Then $\phi(t) = 2V_{\mathrm{dc}}\,t$ and

$$\mathrm{e}^{-\mathrm{i}\phi(t)/2} = \mathrm{e}^{-\mathrm{i}V_{\mathrm{dc}}t} \quad\Longrightarrow\quad \mathcal{T}\,\mathrm{e}^{-\mathrm{i}\phi/2} \text{ is a single harmonic at } \Omega \equiv V_{\mathrm{dc}}. \tag{4}$$

No internal degrees of freedom remain; the current is obtained by a single evaluation of the Green's functions. In the code this corresponds to fixing the harmonic vector to a single nonzero entry, `VipI[2*Nf] = 1` (see `Josephson_Vbias_Floquetn.jl`).

Current bias (the central problem). A fixed dc current is imposed. Only the *time-averaged* voltage $\langle V\rangle$ is fixed; the instantaneous voltage $V(t)$ — equivalently the phase $\phi(t)$ — becomes a dynamical quantity that develops harmonics and must be determined self-consistently. Writing

$$\phi(t) = 2\langle V\rangle\,t + \delta\phi(t), \qquad \delta\phi(t+T_J) = \delta\phi(t), \qquad T_J = \frac{2\pi}{\omega_J}, \quad \omega_J = 2\langle V\rangle, \tag{5}$$

the object that enters the coupling (2) is periodic up to the carrier $\mathrm{e}^{-\mathrm{i}\langle V\rangle t}$. Defining the fundamental Floquet frequency

$$\boxed{\Omega \equiv \langle V\rangle} \tag{6}$$

(this is the variable `Omega` = `ev` in the code), the gauge factor admits the Fourier expansion

$$\mathrm{e}^{-\mathrm{i}\phi(t)/2} \;=\; \sum_{m\ \mathrm{odd}} W_m\,\mathrm{e}^{-\mathrm{i}m\Omega t}, \tag{7}$$

with *only odd harmonics* m present (the carrier $m = 1$ plus the even harmonics of $\delta\phi/2$). The complex coefficients $\{W_m\}$ are the fundamental unknowns of the current-bias problem. Physical observables (the voltage $V(t)$ and the current $I(t)$) are periodic at the Josephson frequency $\omega_J = 2\Omega$, i.e. they contain only *even* harmonics of Ω.

2 Nonequilibrium Green's functions in Nambu space

We work with the contour-ordered Green's functions in the Keldysh representation. For each lead and for the full junction we use the 2×2 Nambu retarded (r), advanced (a) and lesser ($<$) blocks; the advanced block is obtained from the retarded one by Hermitian conjugation, $g^a = (g^r)^\dagger$.

2.1 Surface Green's function of a lead

The local (surface) retarded Green's function of a semi-infinite BCS lead used throughout the code is

$$g^r(\omega) \;=\; \frac{1}{\zeta\,\sqrt{\Delta^2-(\omega+\mathrm{i}\Gamma)^2}}\begin{pmatrix} -(\omega+\mathrm{i}\Gamma) & \Delta \\ \Delta & -(\omega+\mathrm{i}\Gamma)\end{pmatrix} \;=\; \frac{-(\omega+\mathrm{i}\Gamma)\,\tau_0+\Delta\,\tau_1}{\zeta\,\sqrt{\Delta^2-(\omega+\mathrm{i}\Gamma)^2}}, \tag{8}$$

where the branch of the square root is chosen so that $\mathrm{Im}\,g^r \le 0$. This is implemented in `grwmnf` (retarded) and `gawmnf` (advanced); an alternative transfer-matrix construction of the same surface Green's function for a tight-binding chain (following Phys. Rev. B **83**, 085412 (2011)) is provided in `surfacegr`.

The bare lesser function follows from the fluctuation–dissipation relation at zero temperature ($f(\omega) = \theta(-\omega)$):

$$g^<(\omega) = -\left[g^r(\omega) - g^a(\omega)\right]\theta(-\omega) = 2\mathrm{i}\,\theta(-\omega)\ \mathrm{Im}\ g^r(\omega), \tag{9}$$

implemented in `glwmnf` (the step function $\theta(-\omega)$ is applied mode by mode in the Floquet representation below).

2.2 Floquet representation

Because the coupling (2) is periodic in time with fundamental frequency Ω, any two-time function $A(t,t')$ that is periodic in the center-of-mass time can be written in the Floquet (double-Fourier) representation

$$A(t,t') = \sum_{m,n}\int\frac{\mathrm{d}\omega}{2\pi}\,\mathrm{e}^{-\mathrm{i}(\omega+m\Omega)t}\,A_{mn}(\omega)\,\mathrm{e}^{+\mathrm{i}(\omega+n\Omega)t'}, \tag{10}$$

so that products of time-local operators become matrix products over the Floquet indices $m,n \in \{-N_f,\dots,N_f\}$ (the truncation `Nf`). A diagonal (time-local) bare propagator carries shifted energies,

$$\left[g^r(\omega)\right]_{mn} = \delta_{mn}\,g^r(\omega+m\Omega), \tag{11}$$

which is exactly the block-diagonal structure built by `grwmnf`.

The full working space is therefore

$$\underbrace{2}_{L,R}\times\underbrace{2}_{\text{Nambu}}\times\underbrace{(2N_f+1)}_{\text{Floquet}} \;=\; 4(2N_f+1) \tag{12}$$

dimensional. Lead-diagonal quantities are block-diagonal in the L/R index, while the hopping self-energy is purely off-diagonal in L/R.

2.3 Hopping self-energy in Floquet space

Inserting the expansion (7) into the coupling (2), the retarded hopping self-energy becomes a *static* matrix in the Floquet basis that couples the leads. With the Floquet indices m,n labelling rows and columns (code block position $-m+N_f+1$), its lead-off-diagonal Nambu blocks are

$$[\check{V}]^{LR}_{mn} = \mathcal{T}\begin{pmatrix} W_{n-m} & 0\\ 0 & -W^*_{m-n}\end{pmatrix},\qquad [\check{V}]^{RL}_{mn} = \mathcal{T}\begin{pmatrix} W^*_{m-n} & 0\\ 0 & -W_{n-m}\end{pmatrix},\qquad \check{V}^{LL} = \check{V}^{RR} = 0. \tag{13}$$

Here W_r are the Fourier coefficients of $\mathrm{e}^{-\mathrm{i}\phi/2}$ and W_r^* those of $\mathrm{e}^{+\mathrm{i}\phi/2}$ (because ϕ is real), packed in the code into the complex array `Vip` of length $4N_f+1$ whose entries are nonzero only on odd harmonics. This is assembled in `Vwmnf`. The current vertex carries the extra charge matrix τ_3 from the current operator; it is identical to (13) but with the sign of the lower (hole) Nambu entry flipped,

$$\begin{aligned} \check{V}_i \;&=\; (\mathbf{1}_{LR}\otimes\mathbf{1}_{\mathrm{Fl}}\otimes\tau_3)\,\check{V},\\ [\check{V}_i]^{LR}_{mn} = \mathcal{T}\,\mathrm{diag}(W_{n-m},\,+W^*_{m-n}),&\qquad [\check{V}_i]^{RL}_{mn} = \mathcal{T}\,\mathrm{diag}(W^*_{m-n},\,+W_{n-m}), \end{aligned} \tag{14}$$

built by `Viwmnf`. Both $\check{V}$ and $\check{V}_i$ are *linear* in the unknowns $\{W_r\}$ — a fact that is central to the Jacobian construction of §7.

3 Dyson, Keldysh and current equations

3.1 Dyson equation (retarded / advanced)

The full retarded Green's function of the coupled junction follows from the Dyson equation in the $4(2N_f+1)$-dimensional Floquet–Nambu–lead space,

$$\check{G}^r = \big(\mathbf{1} - \check{g}^r\,\check{\Sigma}^r\big)^{-1}\check{g}^r, \qquad \check{G}^a = (\check{G}^r)^\dagger, \tag{15}$$

where $\check{g}^r = \mathrm{diag}(g^r_L, g^r_R)$ is the block-diagonal bare propagator (11) and $\check{\Sigma}^r$ is the hopping self-energy (13). This is `Grwmnf` (solved as a linear system, not by explicit inversion).

3.2 Lesser self-energy

The lesser self-energy collects the occupation information. The code supports two additive contributions (flags `Sigl_b`, `Sigl_s` in `Siglf`):

$$\text{(bath)} \quad \Sigma^<_{\text{bath}}(\omega + m\Omega) = 2\mathrm{i}\,\Gamma\,\theta(-(\omega + m\Omega))\;\tau_0, \tag{16}$$

$$\text{(surface)} \quad \Sigma^<_{\text{surf}}(\omega + m\Omega) = \zeta^2\,\tau_3\,g^<(\omega + m\Omega)\,\tau_3, \tag{17}$$

applied mode by mode. Equation (16) represents local equilibrium reservoirs of strength Γ attached to each site, while (17) embeds the semi-infinite-lead occupation into the contact site.

3.3 Keldysh equation (lesser Green's function)

The full lesser Green's function is obtained from the Keldysh equation

$$\check{G}^< = \check{G}^r\,\check{\Sigma}^<\,\check{G}^a, \tag{18}$$

implemented in `Glesser_Floquet_Tfull`.

3.4 Current formula

The charge current through the junction is the rate of change of charge on one lead, which in terms of the contact self-energy and the lesser propagator reads

$$I(t) = \mathrm{Re}\;\mathrm{tr}\Big[\tau_3\big(\Sigma_{LR}\,G^<_{RL} - \Sigma_{RL}\,G^<_{LR}\big)\Big], \tag{19}$$

where the trace is over Nambu space. In the Floquet representation this becomes a matrix in the Floquet indices, and the harmonic content of the current is

$$I(t) = \sum_h I_{2h}\,\mathrm{e}^{-\mathrm{i}\,2h\Omega t}, \qquad I_{2h} = \frac{1}{2\pi}\int_{-\Omega/2}^{+\Omega/2} \mathrm{d}\omega\;\big[\,\hat{I}(\omega)\,\big]^{\text{harmonic }2h}, \tag{20}$$

where only even harmonics $2h$ of Ω (i.e. harmonics of the Josephson frequency $\omega_J = 2\Omega$) appear. The frequency integral runs over one Floquet Brillouin zone $[-\Omega/2, \Omega/2)$ discretized with spacing `dw0`; the energy grid is `war0`. The dc current is the $h = 0$ component,

$$I_{\text{dc}} = I_0 = \frac{1}{2\pi}\int_{-\Omega/2}^{+\Omega/2} \mathrm{d}\omega\;\mathrm{tr}_{\text{Nambu,Floquet}}\big[\hat{I}(\omega)\big]_{\text{diagonal}}. \tag{21}$$

This is evaluated by `current_Floquet_Tfull` (the Floquet-mode-resolved matrix `Iif`) and summed on the diagonal in `phisolve`. The combination $-\check{V}_i\,\check{G}^<$ in `current_Floquet_Tfull` is exactly (19) written with the τ_3-dressed coupling (13) from `Viwmnf`.

4 Voltage-bias solution

For a constant voltage $V_{\rm dc}$, the phase gauge factor (4) contains a single harmonic, so the coefficient vector reduces to $W_m = \delta_{m,1}$. No self-consistency is required: one simply

1. builds the bare propagators $g^{r,a,<}$ on the grid `war0`;
2. builds the (now trivial) hopping self-energy (13);
3. solves the Dyson equation (15) and Keldysh equation (18);
4. evaluates the current (19).

The dc component $I_{\rm dc}(V_{\rm dc})$ traced out as a function of bias gives the voltage-biased I–V curve. Driver: `Josephson_Vbias_Floquetn.jl`. (A complementary direct real-frequency / multiple-Andreev-reflection evaluation is provided by `Josephson_Vbias_Floquetn_direct.jl`.)

5 Current-bias self-consistency

Under current bias the harmonic coefficients $\{W_m\}$ are unknown and are fixed by three sets of conditions. Let the unknown real vector be the real and imaginary parts of the $2N_f$ odd-harmonic coefficients (so $4N_f$ real unknowns, packed in `Vipi`/`Vipsol`).

(i) Unitarity of the gauge factor. Since $\mathrm{e}^{-\mathrm{i}\phi/2}\,\mathrm{e}^{+\mathrm{i}\phi/2} = 1$ identically, the convolution of the coefficients of $\mathrm{e}^{-\mathrm{i}\phi/2}$ with those of $\mathrm{e}^{+\mathrm{i}\phi/2}$ must equal the unit (zeroth) harmonic and vanish on all others:

$$\sum_k W_k\, W^*_{k-2h} = \delta_{h,0}, \qquad h = 0, \pm 1, \ldots \tag{22}$$

This yields the real and imaginary constraint rows assembled at the top of `IbiasResidual_Tfull` (the products $\mathrm{Re}\,W\ \mathrm{Re}\,W + \mathrm{Im}\,W\ \mathrm{Im}\,W$ etc.), including the normalization $\sum_k |W_k|^2 = 1$.

(ii) Gauge / phase origin. The phase is pinned by $\phi(t{=}0) = 0$, i.e.

$$\sum_m \mathrm{Im}\ W_m = 0, \tag{23}$$

(the row `eqns[2*Nf] = sum(imag.(Vip))`).

(iii) Vanishing of all ac current harmonics (the bias condition). A pure dc current bias means *no* alternating current flows; every nonzero harmonic of the current (20) must vanish:

$$\boxed{I_{2h} = 0 \quad \text{for all } h = 1, \ldots, N_f.} \tag{24}$$

Splitting into real and imaginary parts gives $2N_f$ equations, assembled in `IbiasResidual_Tfull` as

```
eqns[2Nf+h] = Re(Ifa[-2h]),    eqns[2Nf+Nf+h] = Im(Ifa[-2h]).
```

Conditions (22)–(24) together form a square nonlinear system $\mathbf{F}(\mathbf{x}) = \mathbf{0}$ of dimension $4N_f$, where the current harmonics I_{2h} are nonlinear functionals of $\{W_m\}$ through the Dyson/Keldysh equations (15)–(18).

5.1 Numerical solution

The system is solved with a Newton/trust-region method (`NLsolve.nlsolve` with `method=:trust_region`) using the analytically constructed Jacobian (`IbiasJacobian_Tfull`). The bias is swept from high to low $\langle V\rangle$, and the converged coefficients at one bias point seed the solve at the next (numerical continuation). Driver routine: `phisolve` in `Keldyshsetup_Floquetn.jl`; user driver: `Josephson_Ibias_Floquetn.jl`.

Once $\{W_m\}$ are known, the physical voltage waveform is reconstructed from $\phi(t) = 2\mathrm{i}\log(\mathrm{e}^{-\mathrm{i}\phi(t)/2})$ and $V(t) = \frac{1}{2}\dot{\phi}$ (routine `Vt`), and the dc current $I_{\mathrm{dc}}(\langle V\rangle)$ is the current-biased I–V characteristic.

6 Explicit Floquet-component form of the equations

We now write out, component by component, the nonlinear system $\mathbf{F}(\mathbf{x}) = \mathbf{0}$ that `IbiasResidual_Tfull` assembles and `phisolve` solves. This is the precise content of §5.

6.1 The unknown vector

Only odd harmonics of Ω appear in $\mathrm{e}^{-\mathrm{i}\phi/2}$ (7). Index them by $q = 1, \ldots, 2N_f$, corresponding to the odd harmonics

$$r_q = 2q - (2N_f + 1) \in \{-(2N_f - 1), \ldots, -1, +1, \ldots, +(2N_f - 1)\}. \tag{25}$$

The real unknown vector (the array `Vipi`, length $4N_f$) is

$$\mathbf{x} = \big(\underbrace{\mathrm{Re}\,W_{r_1}, \ldots, \mathrm{Re}\,W_{r_{2N_f}}}_{2N_f}, \underbrace{\mathrm{Im}\,W_{r_1}, \ldots, \mathrm{Im}\,W_{r_{2N_f}}}_{2N_f}\big), \qquad W_{r_q} = x_q + \mathrm{i}\,x_{2N_f+q}. \tag{26}$$

6.2 Current in Floquet components

Define the current matrix in the full lead⊗Floquet⊗Nambu space at energy ω,

$$\hat{I}(\omega) \;\equiv\; -\check{V}_i(W)\,\check{G}^<(\omega; W), \tag{27}$$

which is the function `Itemp0=` $-$`Vvi*Glwmn`. Its Floquet-block- and Nambu-resolved trace, taking the LL block minus the RR block, gives the partial current carried between Floquet modes m and n:

$$I_{mn}(\omega) = \mathrm{tr}_{\mathrm{Nambu}}\Big[\hat{I}(\omega)^{LL}_{mn}\Big] - \mathrm{tr}_{\mathrm{Nambu}}\Big[\hat{I}(\omega)^{RR}_{mn}\Big]. \tag{28}$$

Integrating over one Floquet zone $\omega \in [-\Omega/2, \Omega/2)$,

$$I_{mn} = \frac{1}{2\pi}\int_{-\Omega/2}^{\Omega/2} \mathrm{d}\omega\; I_{mn}(\omega) \;\approx\; \frac{\Delta\omega}{2\pi}\sum_{\omega\in\,\texttt{war0}} I_{mn}(\omega), \tag{29}$$

the array `Iif`. The s-th harmonic of the current ($I(t) = \sum_s I_s\,\mathrm{e}^{-\mathrm{i}s\Omega t}$) is the sum over Floquet pairs with fixed index difference,

$$\boxed{I_s \;=\; \sum_{\substack{m,n=-N_f \\ n-m=s}}^{N_f} I_{mn}} \qquad\Longleftrightarrow\qquad \texttt{Ifa}[-(m-n) + (2N_f{+}1)] \mathrel{+}= \texttt{Iif}[\ldots], \tag{30}$$

the array `Ifa`. Because the coupling has only odd harmonics, the current carries only *even* s. The dc current is the diagonal sum

$$I_{\mathrm{dc}} = I_0 = \sum_{m=-N_f}^{N_f} I_{mm} = \mathrm{Re}\ \mathrm{tr}\ \texttt{Iif}. \tag{31}$$

6.3 The residual equations $\mathbf{F}(\mathbf{x})$

The $4N_f$ scalar residuals returned by `IbiasResidual_Tfull` are, in order:

Rows $1\ldots 2N_f-1$ — unitarity/normalization of the gauge factor. Since $\mathrm{e}^{-\mathrm{i}\phi/2}\mathrm{e}^{+\mathrm{i}\phi/2} \equiv 1$, the harmonic-$2h$ component of the product must vanish for $h \neq 0$ and equal unity for $h = 0$. Defining the convolution

$$C_{2h} \;\equiv\; \sum_r W_{r+2h}\, W_r^*, \tag{32}$$

the constraints are

$$\boxed{C_{2h} = \delta_{h,0}} \qquad h = 0, 1, \ldots, N_f - 1, \tag{33}$$

split into

$$F_h = \operatorname{Re} C_{2h} = \sum_r \big(\operatorname{Re} W_{r+2h} \operatorname{Re} W_r + \operatorname{Im} W_{r+2h} \operatorname{Im} W_r\big), \qquad h = 1, \ldots, N_f - 1, \tag{34}$$

$$F_{N_f-1+h} = \operatorname{Im} C_{2h} = \sum_r \big(\operatorname{Im} W_{r+2h} \operatorname{Re} W_r - \operatorname{Re} W_{r+2h} \operatorname{Im} W_r\big), \qquad h = 1, \ldots, N_f - 1, \tag{35}$$

$$F_{2N_f-1} = C_0 - 1 = \sum_r |W_r|^2 - 1. \tag{36}$$

Row $2N_f$ — phase origin. The gauge $\phi(t{=}0) = 0$ makes $\mathrm{e}^{-\mathrm{i}\phi(0)/2} = \sum_r W_r$ real, hence

$$F_{2N_f} = \sum_r \operatorname{Im} W_r = 0. \tag{37}$$

Rows $2N_f + 1 \ldots 4N_f$ — vanishing ac current. A pure dc current bias forbids any ac current; every even harmonic $s = 2h$ with $h = 1, \ldots, N_f$ of (30) must vanish:

$$\boxed{I_{2h}(W) = 0} \qquad h = 1, \ldots, N_f, \qquad \begin{aligned} F_{2N_f+h} &= \operatorname{Re} I_{2h}, \\ F_{3N_f+h} &= \operatorname{Im} I_{2h}. \end{aligned} \tag{38}$$

Equations (34)–(38) are $(2N_f-1)+1+2N_f = 4N_f$ real equations for the $4N_f$ real unknowns (26): a square system. The dc current I_0 is *not* constrained — it is the output read off the converged solution, and sweeping the bias $\Omega = \langle V\rangle$ traces the I–V curve.

7 Newton solution and the analytic Jacobian

7.1 The solver

The square system $\mathbf{F}(\mathbf{x}) = \mathbf{0}$ is solved by `NLsolve.nlsolve` with `method=:trust_region`, supplying both $\mathbf{F}$ (`IbiasResidual_Tfull`) and its exact Jacobian $J_{ij} = \partial F_i/\partial x_j$ (`IbiasJacobian_Tfull`). A trust-region Newton step solves

$$J(\mathbf{x}_k)\,\delta\mathbf{x} = -\,\mathbf{F}(\mathbf{x}_k), \qquad \mathbf{x}_{k+1} = \mathbf{x}_k + \delta\mathbf{x} \tag{39}$$

within an adaptively sized trust region (so divergent Newton steps near the strongly nonlinear gap edges are damped). Convergence uses tolerances `ftol`, `xtol`. The bias is swept from high to low $\langle V\rangle$ and the converged $\mathbf{x}$ at one bias seeds the next solve (numerical continuation), which keeps the iterate inside the basin of attraction across the subharmonic features.

7.2 Jacobian of the algebraic constraints

Rows (34)–(37) are polynomial in $\mathbf{x}$, so their derivatives are written in closed form. Differentiating (32) and using $W_{r_q} = x_q + \mathrm{i}x_{2N_f+q}$:

$$\frac{\partial F_h}{\partial \operatorname{Re} W_q} = \operatorname{Re} W_{q+2h} + \operatorname{Re} W_{q-2h}, \qquad \frac{\partial F_h}{\partial \operatorname{Im} W_q} = \operatorname{Im} W_{q+2h} + \operatorname{Im} W_{q-2h}, \tag{40}$$

$$\frac{\partial F_{N_f-1+h}}{\partial \operatorname{Re} W_q} = \operatorname{Im} W_{q+2h} - \operatorname{Im} W_{q-2h}, \qquad \frac{\partial F_{N_f-1+h}}{\partial \operatorname{Im} W_q} = -\operatorname{Re} W_{q+2h} + \operatorname{Re} W_{q-2h}, \tag{41}$$

$$\frac{\partial F_{2N_f-1}}{\partial \operatorname{Re} W_q} = 2 \operatorname{Re} W_q, \qquad \frac{\partial F_{2N_f-1}}{\partial \operatorname{Im} W_q} = 2 \operatorname{Im} W_q, \tag{42}$$

$$\frac{\partial F_{2N_f}}{\partial \operatorname{Re} W_q} = 0, \qquad \frac{\partial F_{2N_f}}{\partial \operatorname{Im} W_q} = 1, \tag{43}$$

(terms with out-of-range harmonic index are dropped). These are exactly the hand-coded blocks at the top of `IbiasJacobian_Tfull`.

7.3 Jacobian of the current rows

The nontrivial part is $\partial I_{2h}/\partial x_j$. From (27),

$$\frac{\partial \hat{I}}{\partial x_j} = -\frac{\partial \check{V}_i}{\partial x_j}\check{G}^< - \check{V}_i \frac{\partial \check{G}^<}{\partial x_j}. \tag{44}$$

Two simplifications make this cheap and *exact*:

(a) $\check{V}$ and $\check{V}_i$ are linear in $\mathbf{x}$. Hence

$$M_j^{(i)} \equiv \frac{\partial \check{V}_i}{\partial x_j}, \qquad M_j \equiv \frac{\partial \check{V}}{\partial x_j} \tag{45}$$

are *constant*, $\mathbf{x}$-independent, and *sparse*: each unknown x_j (one real or imaginary part of one W_r) sits on a single pair of super/sub-diagonals of the L/R-off-diagonal blocks of (13). The code obtains them once, by a forward finite difference that is exact for a linear map,

$$M_j = \frac{\check{V}(\mathbf{x}+\hat{e}_j) - \check{V}(\mathbf{x})}{\delta V}\bigg|_{\delta V=1}, \qquad M_j^{(i)} = \frac{\check{V}_i(\mathbf{x}+\hat{e}_j) - \check{V}_i(\mathbf{x})}{\delta V}\bigg|_{\delta V=1}, \tag{46}$$

(arrays `Mar`, `Miar`). The thresholding `.* (abs.(Mar) .> 0.5*T*deltaV)` merely discards numerical-zero entries: the genuine nonzero entries all have magnitude $\mathcal{T}$. `Miar` is stored with an overall minus sign ($\texttt{Miar} = -M_j^{(i)}$) to fold the leading minus of (44) into the matrix.

(b) The lesser self-energy does not depend on x. From (16)–(17), $\check{\Sigma}^<$ is built solely from the bare lead propagators and Γ; it contains no W. Therefore $\partial\check{\Sigma}^</\partial x_j = 0$. Differentiating the Dyson equation (15) (with $\check{G}^r = (\mathbf{1} - \check{g}^r\check{V})^{-1}\check{g}^r$, so $\partial\check{G}^r = \check{G}^r M_j \check{G}^r$) and the Keldysh equation (18), and noting that the same hopping matrix $\check{V}$ enters the advanced sector ($\partial\check{G}^a = \check{G}^a M_j \check{G}^a$), gives the compact result

$$\boxed{\frac{\partial \check{G}^<}{\partial x_j} = \check{G}^r M_j \check{G}^< + \check{G}^< M_j \check{G}^a} \tag{47}$$

(the cross terms recombine using $\check{G}^< = \check{G}^r\check{\Sigma}^<\check{G}^a$). Substituting (47) into (44),

$$\frac{\partial \hat{I}}{\partial x_j} = -M_j^{(i)}\check{G}^< - \check{V}_i\Big(\check{G}^r M_j \check{G}^< + \check{G}^< M_j \check{G}^a\Big). \tag{48}$$

This is the exact derivative. Equation (48) is the commented "m0" line in `IbiasJacobian_Tfull`. The line that is actually executed ("m2") is the algebraically identical reordering

$$\frac{\partial \hat{I}}{\partial x_j} = (\underbrace{-\check{V}_i\check{G}^<}_{\texttt{VviGlwmn}})M_j\,\check{G}^a + (\underbrace{-\check{V}_i\check{G}^r}_{\texttt{VviGrwmn}})M_j\,\check{G}^< + (\underbrace{-M_j^{(i)}}_{\texttt{Miar}})\check{G}^<, \tag{49}$$

in which the two factors $-\check{V}_i\check{G}^r$ and $-\check{V}_i\check{G}^<$ are formed *once* per energy ω (outside the loop over the $4N_f$ variables j), leaving only the cheap multiplications by the sparse M_j inside the inner loop. This is the key performance optimization of `IbiasJacobian_Tfull`.

Assembling the current-row Jacobian. For each variable j, the derivative matrix (49) is reduced to Floquet-mode current derivatives exactly as in (28)–(30): take the Nambu trace of the LL minus RR block per Floquet pair (m, n) (array `jacIiwf`), integrate over ω (array `jacIif`), and sum over $n - m = 2h$ (array `jacIfa`). Finally

$$J_{2N_f+h,\,j} = \operatorname{Re}\frac{\partial I_{2h}}{\partial x_j}, \qquad J_{3N_f+h,\,j} = \operatorname{Im}\frac{\partial I_{2h}}{\partial x_j}, \qquad h = 1, \ldots, N_f. \tag{50}$$

7.4 Perturbative Jacobians

For the truncated transparency schemes (`ws` $= 2, 4, 6, 8$; §8) the exact resummed $\check{G}^<$ in (47)–(49) is replaced by its finite expansion in powers of $\check{V}$, built entirely from the *bare* lead propagators $\check{g}^r, \check{g}^a, \check{g}^<$. Because the truncated current is then a finite polynomial in $\mathbf{x}$ (the only $\mathbf{x}$-dependence is through the linear $\check{V}$, $\check{V}_i$), its Jacobian is again *exact* and is obtained by the elementary product rule. This subsection writes out that derivative and the numerical scheme used by `IbiasJacobian_T2/_T4/_T6/_T8`.

Truncated lesser Green's function. Expanding the Dyson/Keldysh equations to finite order ((56)) groups the lesser GF by transparency order,

$$\check{G}^<_{(\le 2P)} = \sum_{p=1}^{P} \check{G}^<_{(2p)}, \qquad \check{G}^<_{(2p)} = \sum_{k=0}^{2p-1} (\check{g}^r\check{V})^k\,\check{g}^<\,(\check{V}\check{g}^a)^{\,2p-1-k}, \tag{51}$$

with $P = 1, 2, 3, 4$ for `ws` $= 2, 4, 6, 8$. Equation (51) is what `Glesser_Floquet_T2/_T4/...` assemble; e.g. at $\mathcal{T}^2$, $\check{G}^<_{(2)} = \check{g}^r\check{V}\check{g}^< + \check{g}^<\check{V}\check{g}^a$, and at $\mathcal{T}^4$ the four chains $k = 0, \ldots, 3$ of $\check{G}^<_{(4)}$ are added.

Derivative of the truncated current. The current operator is $\hat{I} = -\check{V}_i\,\check{G}^<_{(\le 2P)}$, so by the product rule

$$\frac{\partial \hat{I}}{\partial x_j} = \underbrace{-\,M_j^{(i)}\,\check{G}^<_{(\le 2P)}}_{\text{vertex term}}\;\underbrace{-\,\check{V}_i\,\frac{\partial \check{G}^<_{(\le 2P)}}{\partial x_j}}_{\text{propagator term}}, \tag{52}$$

with $M_j^{(i)} = \partial\check{V}_i/\partial x_j$ (array `Miar`, sign-folded) exactly as in §7. Since the bare $\check{g}$'s carry no $\mathbf{x}$, differentiating a single chain in (51) replaces each of its $2p - 1$ factors of $\check{V}$ in turn by $M_j = \partial\check{V}/\partial x_j$ (array `Mar`):

$$\frac{\partial}{\partial x_j}\Big[(\check{g}^r\check{V})^k\check{g}^<(\check{V}\check{g}^a)^{2p-1-k}\Big] = \sum_{\text{each }\check{V}\text{ slot}} \underbrace{(\cdots)}_{L}\;M_j\;\underbrace{(\cdots)}_{R}, \tag{53}$$

i.e. a sum of terms $L\,M_j\,R$ in which L is the sub-chain to the *left* of the replaced $\check{V}$ (ending in a bare $\check{g}$) and R the sub-chain to its *right* (starting with a bare $\check{g}$).

Factorized (computed) form. Multiplying (53) on the left by $-\check{V}_i$ and summing all chains and slots, every contribution to the propagator term of (52) has the form

$$(- \check{V}_i\, L)\; M_j\; R, \tag{54}$$

where the left factors $-\check{V}_i L$ and the right factors R are the *same handful of chains for every variable* j. They are therefore built *once per frequency* and reused across the $4N_f$ columns; only the sparse multiply by M_j sits in the inner loop.

Order $\mathcal{T}^4$ written out. Take $\check{G}^<_{(4)}$, whose $k = 3$ chain is $(\check{g}^r\check{V})^3\check{g}^<$ (three $\check{V}$'s). Replacing each $\check{V}$ and prepending $-\check{V}_i$ gives

$$-\check{V}_i\,\partial_j[(\check{g}^r\check{V})^3\check{g}^<] = \underbrace{(-\check{V}_i\check{g}^r)}_{\texttt{Vvigr}} M_j \underbrace{(\check{g}^r\check{V})^2\check{g}^<}_{\texttt{gr\_s\_gr\_s\_gl}} + \underbrace{(-\check{V}_i\check{g}^r\check{V}\check{g}^r)}_{\texttt{Vvigr\_s\_gr}} M_j \underbrace{\check{g}^r\check{V}\check{g}^<}_{\texttt{gr\_s\_gl}} + \underbrace{(-\check{V}_i(\check{g}^r\check{V})^2\check{g}^r)}_{\texttt{Vvigr\_s\_gr\_s\_gr}} M_j \underbrace{\check{g}^<}_{\texttt{glwmn}}, \tag{55}$$

which is exactly one line of the kernel in `IbiasJacobian_T4`. Here the naming convention is: `s` denotes the hopping $\check{V}$ and `r/l/a` denote $\check{g}^r/\check{g}^</\check{g}^a$, so `gr_s_gl`$= \check{g}^r\check{V}\check{g}^<$ and `Vvigr_s_gr`$= -\check{V}_i\,\check{g}^r\check{V}\check{g}^r$. The other three chains ($k = 2, 1, 0$, ending in $\check{g}^a$ factors) contribute the remaining lines; together with the $\mathcal{T}^2$ pair $-\check{V}_i\check{g}^r M_j\check{g}^<$, $-\check{V}_i\check{g}^< M_j\check{g}^a$ and the vertex term $-M_j^{(i)}\check{G}^<_{(4)}$ (code `Miar*Glwmn_w4`) this is the full $\partial\hat{I}/\partial x_j$ at $\mathcal{T}^4$ — 15 matrix products plus the vertex term. At $\mathcal{T}^6, \mathcal{T}^8$ the same construction adds the order $\mathcal{T}^6, \mathcal{T}^8$ chains; the kernels group them as `t3_1` ($\mathcal{T}^2$), `t3_3` ($\mathcal{T}^4$), `t3_5` ($\mathcal{T}^6$), `t3_7` ($\mathcal{T}^8$) and `t3_f` (vertex term).

Numerical manipulations. For a given bias the routine performs, for each frequency $\omega \in$ `war0`:

1. build the bare propagators $\check{g}^r$ (`grwmn`), $\check{g}^a = \check{g}^{r\dagger}$ (`gawmn`), $\check{g}^<$ (`glwmn`), and the truncated $\check{G}^<_{(\leq 2P)}$ (`Glwmn_wN` via `Glesser_Floquet_TN`);
2. build the right chains R — e.g. `gr_s_gl`, `gl_s_ga`, `gr_s_gr_s_gl`, … — by repeated multiplication by $\check{V}$ (`Vv`);
3. build the left chains $-\check{V}_i L$ — `Vvigr`, `Vvigr_s_gr`, … — by one `mul!` each (*computed once*, reused for all j; at $\mathcal{T}^6/\mathcal{T}^8$ the buffers are overwritten in place with `.=`);
4. loop (threaded) over the $4N_f$ variables j: assemble the derivative matrix as the order-grouped sum `t3` = `t3_1` + `t3_3` + … of the terms (54) with the sparse M_j (`Mar[:,:,j]`) plus the vertex term `Miar[:,:,j]*Glwmn_wN`;
5. reduce `t3` to the Floquet-mode current derivative by the $LL - RR$ Nambu trace and accumulate the frequency integral on the fly into `jacIif` (Tier-2 accumulation of §7).

Finally `jacIif` is scaled by $\Delta\omega/2\pi$, collapsed onto current harmonics (`jacIfa`, summing $n - m = 2h$), and written to the current rows (50).

Two implementation notes. (i) `IbiasJacobian_T2` does not need the chain machinery: $\partial_j\check{G}^<_{(2)} = \check{g}^r M_j\check{g}^< + \check{g}^< M_j\check{g}^a$ has the same three-term shape as the exact case, so it reuses the sparse-reordered form (49) verbatim but with bare propagators and $\check{G}^<_{(2)}$ in the vertex term. (ii) `IbiasJacobian_T6/_T8` are provided for completeness; `phisolve` currently runs `ws` $= 6, 8$ *without* supplying an analytic Jacobian (`NLsolve` then finite- differences the residual), and the $\mathcal{T}^6$ kernel omits the vertex term that the $\mathcal{T}^8$ kernel keeps (`t3_f`).

8 Perturbative (transparency) expansion

To dissect which microscopic processes produce each feature of the I–V curve, the lesser Green's function is also evaluated as a power series in the transparency (a Werthamer-type tunnelling expansion). Expanding the Dyson equation (15) and Keldysh equation (18) in powers of $\check{\Sigma}^r \propto \mathcal{T}$,

$$\check{G}^< = \sum_{p\geq 1} \sum_{k=0}^{2p-1} (\check{g}^r \check{\Sigma}^r)^k \, \check{g}^< \, (\check{\Sigma}^r \check{g}^a)^{2p-1-k}, \tag{56}$$

and truncating at order $\mathcal{T}^2$, $\mathcal{T}^4$, $\mathcal{T}^6$, $\mathcal{T}^8$ gives the functions `Glesser_Floquet_T2`, `_T4`, `_T6`, `_T8` respectively (and the corresponding currents `current_Floquet_T2`..., self-consistency equations `IbiasResidual_T2`...). The order is selected by the `ws` flag (`ws` $= 0$ is the exact Dyson resummation; `ws` $= 2, 4, 6, 8$ the truncated orders). The lowest nontrivial order $\mathcal{T}^2$ is the single-quasiparticle (and lowest pair) contribution; the $\mathcal{T}^4$ order is where the *two-quasiparticle* processes that the paper identifies as the origin of the current-biased subharmonic gap structure first appear.

9 Observables and the subharmonic gap structure

The principal outputs are:

- the dc I–V curve $I_{\mathrm{dc}}(\langle V\rangle)$, usually plotted in the dimensionless form $I\, eR_N/\Delta$ versus eV/Δ;
- the differential conductance $\mathrm{d}I/\mathrm{d}V$, in which the subharmonic gap structure (SGS) appears as a series of features at

$$eV = \frac{2\Delta}{n}, \qquad n = 1, 2, 3, \ldots \tag{57}$$

  (the dashed vertical lines in the plotting code).

The normal-state resistance R_N used for normalization is obtained from the same machinery by setting $\Delta = 0$ and reading off the linear (Ohmic) response, $R_N = \langle V\rangle / I_{\mathrm{dc}}|_{\Delta=0}$ (routine `current_full_RN`).

10 Extension to the 4×4 Nambu⊗spin basis: classical-spin (YSR) impurities

All of the preceding development uses a 2×2 Nambu space, which is sufficient for a clean spin-singlet junction. To host *classical-spin (magnetic) impurities* — and thereby Yu–Shiba–Rusinov (YSR) bound states and the Josephson diode effect — the spin degree of freedom must be made explicit. This section documents the 4×4 Nambu⊗spin extension implemented in `Keldyshsetup_Floquetn_ext.jl`, following Eqs. (7)–(8) of arXiv:2602.15213. The structure of §§2–7 is reused verbatim; we highlight only what changes.

10.1 Basis and the spin-resolved surface Green's function

The Nambu spinor $\Psi = (c_\uparrow, c_\downarrow^\dagger)^{\mathsf{T}}$ of §1 is replaced by the full Nambu⊗spin spinor

$$\check{c} \;=\; \big(c_\uparrow,\, c_\downarrow,\, c_\uparrow^\dagger,\, c_\downarrow^\dagger\big)^{\mathsf{T}} \;=\; \tau \otimes \sigma, \tag{58}$$

with τ the particle–hole and σ the spin structure (particle block $(\uparrow,\downarrow)$, hole block $(\uparrow^\dagger,\downarrow^\dagger)$). The clean BCS surface Green's function (8) becomes the 4×4 matrix

$$\check{g}_0(z) \;=\; \frac{1}{\zeta\sqrt{\Delta^2-z^2}}\begin{pmatrix}-z & 0 & 0 & \Delta\\ 0 & -z & -\Delta & 0\\ 0 & -\Delta & -z & 0\\ \Delta & 0 & 0 & -z\end{pmatrix}, \qquad z=\omega+\mathrm{i}\Gamma, \tag{59}$$

in which the singlet pairing couples $c_\uparrow \leftrightarrow c_\downarrow^\dagger$ with $+\Delta$ (*block A*) and $c_\downarrow \leftrightarrow c_\uparrow^\dagger$ with $-\Delta$ (*block B*). Equation (59) is just the two time-reversed copies of (8) embedded with opposite-sign gaps; it is the helper `gsurf4` evaluated at $\check{V}_{\mathrm{imp}}=0$.

10.2 Classical-spin impurity and the local Dyson dressing

A classical impurity spin (exchange $\mathbf{J}=(J_x,J_y,J_z)$, potential K) enters as a static, local self-energy [Eq. (7) of arXiv:2602.15213]

$$\check{V}_{\mathrm{imp}} \;=\; \sum_{k=x,y,z} J_k\,(\tau_u\otimes\sigma_k-\tau_d\otimes\sigma_k^*) \;+\; K\,(\tau_z\otimes\sigma_0) \;=\; \begin{pmatrix}\mathbf{J}\cdot\boldsymbol{\sigma} & 0\\ 0 & -(\mathbf{J}\cdot\boldsymbol{\sigma})^*\end{pmatrix}+K\begin{pmatrix}\sigma_0 & 0\\ 0 & -\sigma_0\end{pmatrix}, \tag{60}$$

with the particle/hole projectors $\tau_u=\frac{1}{2}(\tau_0+\tau_z)$, $\tau_d=\frac{1}{2}(\tau_0-\tau_z)$ (helper `impurity4`). The longitudinal part J_z is diagonal in spin and keeps blocks A and B decoupled (it only shifts them oppositely, splitting the YSR levels); the transverse parts J_x, J_y are off-diagonal in spin and *couple* A and B.

The impurity is resummed into the surface propagator by a local Dyson (T-matrix) dressing,

$$\boxed{\check{g}(z) \;=\; \big(\mathbf{1}_4-\check{g}_0(z)\,\check{V}_{\mathrm{imp}}\big)^{-1}\check{g}_0(z)} \tag{61}$$

(the body of `gsurf4`), whose in-gap poles are the YSR bound states. Each lead is dressed *independently*: the left lead uses $(\mathbf{J}_L,K_L)$ and the right lead $(\mathbf{J}_R,K_R)$, so the two contacts are in general inequivalent (code arguments `JL,KL,JR,KR`).

10.3 Floquet / lead layout

The Floquet construction of §2.2 is unchanged except for the larger inner block. The bare lead propagators are block-diagonal in the Floquet index,

$$[\check{g}^r]_{mn}=\delta_{mn}\,\check{g}(\omega+m\Omega+\mathrm{i}\Gamma), \tag{62}$$

with $\check{g}$ from (61), and the lesser function $\check{g}^< = -(\check{g}^r-\check{g}^a)\,\theta\big(-(\omega+m\Omega)\big)$ as in (9) — now automatically including the equilibrium filling of any YSR level. The working space grows to

$$\underbrace{2}_{L,R}\times\underbrace{4}_{\text{Nambu}\otimes\text{spin}}\times(2N_f+1) \;=\; 8(2N_f+1), \tag{63}$$

versus $4(2N_f+1)$ in the 2×2 theory. Routines: `grwmnf`, `gawmnf`, `glwmnf` (now taking `JL,KL,JR,KR`).

10.4 Hopping and current vertex are spin-conserving

The inter-lead tunnelling [Eq. (8) of arXiv:2602.15213] conserves spin, $\mathcal{T}\,(W\,\tau_u-W^*\tau_d)\otimes\sigma_0$, so the Floquet hopping self-energy (13) is simply the 2×2 Nambu matrix tensored with the spin identity,

$$[\check{V}]^{LR}_{mn}=\mathcal{T}\begin{pmatrix}W_{n-m} & 0\\ 0 & -W^*_{m-n}\end{pmatrix}\otimes\sigma_0, \qquad [\check{V}_i]^{LR}_{mn}=\mathcal{T}\begin{pmatrix}W_{n-m} & 0\\ 0 & +W^*_{m-n}\end{pmatrix}\otimes\sigma_0, \tag{64}$$

and likewise for the RL blocks. In particular $\check{V}$ and $\check{V}_i$ carry *no* impurity parameters (these live only in the leads), so `Vwmnf/Viwmnf` keep their 2×2-era signatures, differing only by the `kron(.,sig0)`. They remain *linear* in $\{W_r\}$, which is what preserves the entire Jacobian construction below.

10.5 Dyson, Keldysh, and the current

The Dyson equation (15), the lesser self-energy (16)–(17) (with $\tau_0 \to \mathbf{1}_4$ and $\tau_3 \to \tau_z \otimes \sigma_0$), and the Keldysh equation (18) are formally identical, now solved in the $8(2N_f+1)$-dimensional space (`Grwmnf`, `Siglf`, `Glesser_Floquet_Tfull`). The current (19) is unchanged in form, $\hat{I} = -\check{V}_i\check{G}^<$, but the Nambu trace becomes a Nambu$\otimes$spin trace over *four* diagonal components, with the right lead at offset $4(2N_f+1)$:

$$I_{mn}(\omega) = \sum_{a=1}^{4} [\hat{I}^{LL}_{mn}]_{aa} - \sum_{a=1}^{4} [\hat{I}^{RR}_{mn}]_{aa}, \tag{65}$$

to be compared with the two-component trace (28). Everything downstream — the frequency integral (29), the harmonic collapse (30), the dc current — is identical (routine `current_Floquet_Tfull`, and the perturbative `current_Floquet_T2/_T4`).

10.6 Self-consistency and Jacobian: what changes and what does not

The phase $\phi(t)$ is a *scalar* (spin-independent) gauge field, so the unknown vector (26) is *unchanged*: the same $4N_f$ reals. Consequently, in the residual $\mathbf{F}(\mathbf{x})$ of §6,

- the unitarity rows (34)–(36) and the gauge row (37) are *identical* (they involve only $\{W_r\}$, not the basis);
- only the ac-current rows (38) change — through the 4×4 current (65) and the per-lead impurity.

The analytic Jacobian inherits the same split. The constraint-row derivatives (40)–(43) are unchanged. For the current rows the two key facts of §7 still hold verbatim: $\check{\Sigma}^<$ is independent of $\mathbf{x}$, and $\check{V}, \check{V}_i$ are linear in $\mathbf{x}$, so

$$\frac{\partial \check{G}^<}{\partial x_j} = \check{G}^r M_j \check{G}^< + \check{G}^< M_j \check{G}^a, \qquad \frac{\partial \hat{I}}{\partial x_j} = (-\check{V}_i\check{G}^<)M_j\check{G}^a + (-\check{V}_i\check{G}^r)M_j\check{G}^< + (-M_j^{(i)})\check{G}^<, \tag{66}$$

exactly Eqs. (47) and (49). The *only* modifications are dimensional: the constant sparse derivatives $M_j, M_j^{(i)}$ and all scratch matrices grow from $4(2N_f+1)$ to $8(2N_f+1)$, and the per-Floquet-pair reduction uses the four-component Nambu$\otimes$spin trace (65). The perturbative chain derivatives of §7.4 carry over identically (wider matrices, four-component trace). Routines: `IbiasResidual_Tfull`, `IbiasJacobian_Tfull`, and the truncated `_T2/_T4`; `phisolve` dispatches `ws` $= 0, 2, 4$ with `JL,KL,JR,KR` threaded through.

10.7 Reduction to the 2×2 theory, and the diode effect

For a non-magnetic contact ($\mathbf{J}_L = \mathbf{J}_R = 0$, $K_L = K_R = 0$) the dressing (61) is trivial and $\check{g}_0$ (59) is block-diagonal into block A — which is *precisely* the original 2×2 problem (8) — and its time-reverse block B. Because the hopping (64) is $\propto \sigma_0$, the two blocks never mix, and the four-component trace (65) equals twice the single-block trace. Hence every observable computed in the extended module reduces to *exactly twice* its 2×2 counterpart — the spin-degeneracy factor. This is the regression used to validate the port: with $\mathbf{J} = K = 0$, `current_Floquet_T2/_T4` and the current rows of `IbiasResidual`/`IbiasJacobian` reproduce $2\times$ the `Keldyshsetup_Floquetn.jl` values to machine precision, and the analytic Jacobian matches a finite difference of the residual.

The physics that genuinely requires the 4×4 machinery is the *non-reciprocal* (diode) regime: for non-collinear or unequal leads ($\mathbf{J}_L \nparallel \mathbf{J}_R$) blocks A and B are coupled and the left/right symmetry is broken, $I_{LR} \neq -I_{RL}$, so the supercurrent is no longer odd in the phase/voltage — the Josephson diode effect that motivates the extension.

10.8 Correspondence to the extended code

Equation / object	**Routine in `Keldyshsetup_Floquetn_ext.jl`**
Clean 4×4 surface GF (59)	`gsurf4` (with $\check{V}_{\mathrm{imp}} = 0$)
Impurity self-energy $\check{V}_{\mathrm{imp}}$ (60)	`impurity4(J,K)`
Local Dyson dressing (61)	`gsurf4(z,zeta,delta,Vimp)`
Dressed lead GFs $g^{r,a,<}$	`grwmnf`, `gawmnf`, `glwmnf` (`...,JL,KL,JR,KR`)
Hopping / current vertex (64)	`Vwmnf`, `Viwmnf` (`kron(.,sig0)`)
Dyson / lesser SE / Keldysh	`Grwmnf`, `Siglf`, `Glesser_Floquet_Tfull`
Current, four-component trace (65)	`current_Floquet_Tfull`
Transparency expansion ($\mathcal{T}^2, \mathcal{T}^4$)	`Glesser/current_Floquet/IbiasResidual/IbiasJacobian_T2,`
Self-consistency / Jacobian (66)	`IbiasResidual_Tfull`, `IbiasJacobian_Tfull`
Bias sweep + Newton solve (`ws` $= 0, 2, 4$)	`phisolve(...,JL,KL,JR,KR)`
Normal-state resistance R_N	`RN_full(...,JL,KL,JR,KR)`

What differs from the 2×2 routines. The signatures gain the per-lead impurity arguments `JL,KL,JR,KR`; matrices are sized $8(2N_f + 1)$; the Nambu trace sums four diagonal components with right-lead offset $4(2N_f + 1)$; and `Vwmnf`/`Viwmnf` tensor the 2×2 hopping with σ_0. All other differential and algebraic structure is identical to §§6–7.

11 Correspondence to the code

Equation / object	**Routine in `Keldyshsetup_Floquetn.jl`**
Surface GF g^r, g^a (8)	`grwmnf`, `gawmnf`
Surface GF (transfer matrix)	`surfacegr`
Bare lesser $g^<$ (9)	`glwmnf`
Hopping self-energy $\Sigma_{LR/RL}$ (13)	`Vwmnf`, `Viwmnf`
Dyson eq. $\check{G}^r$ (15)	`Grwmnf`, `Gawmnf`
Lesser self-energy (16), (17)	`Siglf`
Keldysh eq. $\check{G}^<$ (18)	`Glesser_Floquet_Tfull`
Transparency expansion (56)	`Glesser_Floquet_T2/_T4/_T6/_T8`
Current (19), (20)	`current_Floquet_Tfull`, `current_Floquet_T2/...`
Self-consistency $\mathbf{F}(\mathbf{x})$ (22)–(24)	`IbiasResidual_Tfull`, `IbiasResidual_T2/...`
Jacobian of $\mathbf{F}$	`IbiasJacobian_Tfull`, `IbiasJacobian_T2/...`
Bias sweep + Newton solve	`phisolve`
Voltage waveform $V(t)$	`Vt`
Normal-state resistance R_N	`current_full_RN`

Index conventions. Matrices act on the ordered product space (lead)$\otimes$(Floquet)$\otimes$(Nambu). A Floquet index $m \in \{-N_f, \dots, N_f\}$ maps to the block row/column $(-m+N_f+1)$; the complex coefficient array `Vip` of length $4N_f+1$ is centered at index $2N_f+1$ ($\leftrightarrow m=0$) and is nonzero only on odd m. The solver's real unknown vector `Vipi`/`Vipsol` stacks the $2N_f$ real parts followed by the $2N_f$ imaginary parts of these odd harmonics.

References

1. *Origin of Subharmonic Gap Structure of DC Current-Biased Josephson Junctions*, arXiv:2412.09862.

2. Surface Green's function / transfer-matrix construction: Phys. Rev. B **83**, 085412 (2011).